One Truck's Journey — Today

AxlEye + Mettler Toledo DataBridge MS

Left to right: arrival → permit → measurement → ticketing → loading → outbound. Time roughly 3-5 minutes from arrival to InYard. Color shows who owns the step. **Note:** FastWeigh is NOT installed today — Jahna's DataBridge→FastWeigh migration happens in 2-3 weeks, concurrent with our AxlEye replacement.

Driver / Operator | AxlEye / Mettler Toledo DataBridge MS (going away) | FDOT (regulator)

1

DRIVER

Truck onto inbound scale

Drop axle deployed. Driver holds OW permit up to driver-window camera.

2

AXLEYE

QR + FDOT lookup

Reads QR, calls FDOT public PAS URL, saves permit JPG to scale-PC folder.

3

AXLEYE + OPERATOR

Measure axles

Operator marks first / last axle lines. Bridge formula calculates max gross. XML + JPG saved to folder.

4

DATABRIDGE PLUGINS

AxlEye + file-rename plugins

DataBridge's AxlEye plugin pulls XML into the ticket. Second plugin renames XML + JPG with ticket # + timestamp once ticket exists.

5

METTLER TOLEDO DATABRIDGE MS + OPERATOR

Ticket created

Operator selects hauler / truck / customer, "Get Weight", "Accept". DataBridge reads IND780 indicator, creates ticket in local SQL Server. InYard.

6

CUSTOM HTML POLLER

Loader sees max weight

Customer-written HTML script on tablet polls scale-PC folder, renders InYard truck + max weight target.

7

DATABRIDGE MS + OPERATOR

Outbound & complete

Truck loaded, returns to outbound scale. Operator "Complete" in DataBridge. Ticket prints via Crystal Reports, some customers emailed.

Today's audit trail is filesystem-based and per-site. There is no centralized cross-site view of permits, measurements, or alerts. AxlEye + DataBridge are tightly coupled — DataBridge MS owns ticketing and a DataBridge plugin orchestrates AxlEye's measurement output. Jahna's migration from DataBridge to FastWeigh is independently scheduled for the same 2-3 week window as our AxlEye replacement.

One Truck's Journey — New

Dividia NVR + Jetson + FastWeigh

Same operator workflow, our hardware + software underneath. Manual line placement preserved for day one. AI auto-detect is a Phase 2 software-only upgrade on the same Jetson.

Driver / Operator | Dividia (new) | FastWeigh | FDOT (regulator)

1

DRIVER

Truck onto inbound scale

Scale-on event fires from IP scale. Drop axle deployed. Driver holds permit up to QR cam.

2

DIVIDIA RECORDER

Transaction opens

Scale-on triggers a fresh TruckTransaction. Recorder is now the source of truth for what this truck did at this site.

3

QR CAMERA (REUSED)

Permit QR decoded

Recorder's QR thread decodes pas.fdot.gov URL off the frame. Emits `Qr_Read` event into the open transaction.

4

FDOT PAS (PUBLIC URL)

Permit PDF fetched

Recorder calls public PAS URL, downloads PDF, parses permittee + expiration + axle limits. **No credentials.**

5

JETSON SPOT DECODER + OPERATOR

Measure axles

Side-view on HDMI. Operator turns the knobs on their existing Razer Stream Controller to place the lines. Bridge formula + permit caps → max gross. Phase 2: AI marks lines, operator just reviews.

6

FASTWEIGH + OPERATOR

Ticket created → webhook

Operator's normal FastWeigh entry. FastWeigh fires `ticket.created` webhook to Dividia cloudapi.

7

CLOUDAPI → RECORDER

Ticket joins transaction

cloudapi pushes the ticket via WebSocket. Recorder merges into the open transaction (which already has scale + QR + measurement + permit).

8

SCALEWATCHER PORTAL

Audit trail + cross-site view

Full transaction record: permit PDF, measurement image, scale weights, alerts. Searchable across all 6 sites. New capability Jahna does not have today.

Same operator clicks, same camera positions, same physical workflow. The replumbing is underneath. Manual line placement is preserved at Phase 1; Phase 2 is a software-only upgrade on the same Jetson.

Where the Two Flows Differ

Five moments that drive the quote

For each moment, what happens today vs what happens after the switch. Read this when sales / ops ask "so what actually changes?"

Moment	Today (AxlEye + DataBridge)	New (Dividia + FastWeigh)
1. Transaction starts when …	**Loose.** AxlEye runs *before* the DataBridge ticket exists. There is no unified "transaction" — just file artifacts that DataBridge's AxlEye plugin stitches together by ticket number afterward. If the operator skips a piece, the linkage breaks silently.	**Scale-on.** The IP scale firing its first reading opens a TruckTransaction inside our recorder. Everything that happens — QR scan, permit, measurement, FastWeigh ticket — gets attached to that open transaction by time and lane, not by guesswork.
2. Permit reading lives in …	AxlEye reads QR + downloads permit JPG to a folder on the scale PC. **If AxlEye stops running, permit handling stops with it.** The XML it produces carries a TransactionGuid that DataBridge's AxlEye plugin uses to attach measurement to the right ticket.	Recorder reads QR off the camera frame, calls FDOT's *public* permit URL, fetches the PDF, parses permittee / expiration / axle caps. No credentials. The PDF rides into our cloud as an artifact, attached to the same TruckTransaction. Full chain of custody.
3. Axle measurement runs on …	AxlEye PC (~$3K hardware, end-of-life vendor) with a wired monitor. Operator marks lines on the AxlEye UI. Output: XML + JPG file pair, dropped on disk. **The vendor is going out of business; the install base is 5 sites total.**	Jetson Orin Nano (Dividia "Spot Decoder") running our C code. Operator uses the same Razer Stream Controller they have today (USB, knobs encode line position) — no retraining on a new input device. Bridge formula calculation, INI persistence, HTTP API to recorder. **Phase 2 (software-only upgrade on the same Jetson):** AI auto-detects first / last axle; operator becomes reviewer.
4. Loader operator sees max weight via …	Customer-written HTML script polling the scale-PC folder. Renders InYard trucks + max-weight target to each tablet. Brittle — depends on AxlEye + DataBridge both running and the folder being accessible from the tablets' network.	FastWeigh's loader-tablet app surfaces the ticket. **Open question for Brad (see FastWeigh Call Bullets):** does FastWeigh's app render our external measurement / allowed-gross? If yes, our number reaches the operator natively. If no, we keep the HTML poller running against our new output folder as a stopgap, or build a Dividia tablet display.
5. Audit / cross-site visibility …	**None.** Each site is an island. Permit images, measurement images, ticket data live in different places. There is no "show me every overweight alert across all 6 sites this week." DOT auditor visit means hand-pulling files.	ScaleWatcher portal aggregates every truck across every site. Search by permit number, ticket, alert type, date range. Permit PDF + measurement image always one click away. New capability that didn't exist before — and it is the *constant* across every reliability and automation tier we quote.
6. What survives a lightning hit on …	Whatever AxlEye PC or DataBridge PC was doing at the strike moment is lost. Both PCs are single points of failure with no spare strategy. AxlEye vendor is dormant — replacement is uncertain. DataBridge is being phased out anyway as part of Jahna's FastWeigh migration.	**Standard tier:** UPS + surge protection + pre-imaged recovery USBs + heartbeat alerts. **Failover tier:** hot-spare Jetson + cold-spare NVR + LTE backup uplink + reconciliation worker for missed cloud webhooks. Non-technical operator can swap a Jetson in under 5 minutes following the printed runbook.

For the deep-dive on each moment, see the companion documents: *Current vs Future Flow* for full step-by-step + equipment BOM; *Resilience & Recovery* for failure modes + swap playbook; *FastWeigh Call Bullets* for the conditional that controls moment #4 above.